HEARTGUARD

AI-Based Heart Disease Risk Assessment Report · Academic / Research Prototype

Assessment ID	21dd74a4	**Assessment Date**	2026-09-08 19:19:50
Patient ID	UID-0005	**Model Version**	logistic_regression (v1.0.0)
Alert Status	NOT_TRIGGERED	**Evaluation Type**	Multimodal (70% ML + 30% NLP)

55.9%

OVERALL MULTIMODAL RISK SCORE

MONITORING

Recommended Action:
Regular monitoring recommended.

Assessment Narrative

HeartGuard estimated a clinical model risk of 64.9% and a lifestyle risk score of 35/100. Using the configured 70/30 multimodal weighting (70% clinical ML + 30% lifestyle NLP), the resulting overall risk is 55.9%. Assessment Classification: MONITORING. Primary lifestyle signals detected: Smoking and alcohol use. Recommended action: Regular monitoring recommended. Disclaimer: HeartGuard is an academic/research prototype and is not a medical diagnostic system. The overall score is an experimental model-based risk assessment and should not replace evaluation by a qualified healthcare professional.

Risk Components & Explainability

HeartGuard decomposes cardiovascular risk into objective clinical measurements and qualitative lifestyle habits.

Modality	Raw Score	Weight	Weighted Contribution
Clinical ML (Physiological Metrics)	64.9%	70%	+45.43 pts
Lifestyle NLP (Habits & Narratives)	35.0%	30%	+10.50 pts
Overall Combined Risk	55.9%	100%	55.93 pts

Top Clinical Risk Drivers (SHAP Analysis)

Rank	Clinical Parameter	SHAP Impact Score
1	Chest Pain Type 3	+0.3950
2	Chest Pain Type 1	+0.1139
3	Exercise Angina Absent	+0.1118

Detected Lifestyle Signals

Factor	Severity	Risk Impact
Smoking	HIGH	+25 pts
Alcohol Use	LOW	+10 pts

Historical Context & Risk Trajectory

Evaluation of how model predictions compare against prior assessments.

Single assessment record available — comparative change metrics require at least two historical assessments.

Risk Trend Trajectory: Insufficient historical data for trend analysis. (Minimum 2 assessments required).

AI Insights & Personalized Recommendations

Explainable, action-oriented guidance derived deterministically from your multimodal assessment data.

Personalized Assessment Summary

Your latest HeartGuard assessment indicates a model-based risk score of 55.9% (Monitoring). Features with the highest statistical influence on this output included Chest Pain Type 3, Chest Pain Type 1 and Exercise Angina Absent.

Top Model Contributors (Explainability)

• **Chest Pain Type 3:** Chest Pain Type 3 contributed to this model-based assessment score based on statistical feature importance.
• **Chest Pain Type 1:** Chest Pain Type 1 contributed to this model-based assessment score based on statistical feature importance.
• **Exercise Angina Absent:** Exercise Angina Absent contributed to this model-based assessment score based on statistical feature importance.

Personalized Recommendations

Priority	Category	Recommended Guidance
HIGH	Smoking	**Consider exploring smoking cessation support** Smoking and tobacco use was identified as an elevated lifestyle risk factor in your assessment. Consider speaking with a healthcare professional about evidence-based cessation strategies and community health resources.
LOW	Alcohol	**Consider moderating alcohol consumption** Alcohol intake was identified in your lifestyle inputs. Moderating intake in accordance with standard cardiovascular health guidelines supports long-term wellness.
LOW	Assessment Monitoring	**Maintain current healthy preventive routines** Your assessment falls within the lower baseline risk tier. Continue maintaining balanced daily physical activity, nutritious eating, and regular routine wellness visits.

Notice: *These AI-generated insights are informational and are not a medical diagnosis or treatment plan.*

Alert Information & Disclaimers

Alert Dispatch Status	**NOT_TRIGGERED**
Privacy Protection	Recipient phone numbers are masked in all audit trails and reports (e.g. +1*****4567) to ensure patient confidentiality.
Notification Note	Emergency alerts are triggered strictly when overall risk exceeds the critical threshold (> 85%).

Medical & Research Disclaimers

Academic & Research Prototype Notice:
HeartGuard is an academic/research prototype for AI-based heart disease risk assessment. It is not a medical diagnostic system and does not replace evaluation by a qualified healthcare professional. Model predictions are experimental computational estimates.

Risk Score Interpretation:
Changes in HeartGuard risk scores represent changes in the model's assessment inputs and outputs. They do not by themselves establish a medical diagnosis or disease progression. HeartGuard does not make clinical diagnoses, prescribe medication, or formulate treatment plans. Always seek the counsel of a physician or other qualified health provider with any questions regarding a medical condition.